

BIS SATHI — AI Assistant for Indian Standards & BIS Services

Problem Statement ID: SIH26107 | Theme: Smart Automation | Category: Software

Team: CONTEXT CREW | Live Production: <https://bis-agnet.app/> | Evaluation Date: September 2026

EXECUTIVE MASTER BRIEFING

BIS SATHI (Bureau of Indian Standards Smart Automated Technical Help Interface) is a production-deployed, multilingual Agentic RAG system built to bridge the gap between 23,000+ Indian Standards (IS), 1,437+ certified product categories, and 5.70 Crore Indian MSMEs and consumers. This comprehensive dossier combines rigorous technical architecture analysis, comparative findings from the hackathon submissions, verbatim slide-by-slide winning pitch speeches, live demo playbooks, and strategic Jury defense battlecards.

1. Comparative Analysis of Submitted Hackathon Documents

A synthesis was performed comparing the two foundational hackathon submissions: **Document 1 (SIH 2026 6-Slide Presentation Deck)** and **Document 2 (Technical Executive Summary & Whitepaper)**. Both documents represent a unified vision designed to solve the complexity and fragmentation of the Indian regulatory ecosystem.

Document Section	Document 1: 6-Slide Pitch Deck	Document 2: Technical Whitepaper	Codebase Realization & Status
Problem Definition	Highlights scattered info, 23K+ standards, MSME compliance costs, and lack of guidance across portals.	Identifies 5 key bottlenecks: scattered data, technical jargon, procedural opacity, search delay, and accessibility.	Solved via multi-intent routing, conversational memory, and plain-language synthesis.
System Workflow	Features the dual-tier visual workflow: 0-Token Fast Path (₹0.00) vs. Smart AI Path.	Details the exact procedural steps: OAuth Gateway -> Cache Lookup -> Hybrid Search -> Live Portal Scraper -> Synthesis.	Implemented in src/agent/graph.py and src/agent/session.py.
Technology Stack	Visual logos: React, FastAPI, LangGraph, Whisper, Gemini, Mistral, PGVector, Supabase, Vercel, Render.	Structured component table pairing each layer with exact software packages and frameworks.	100% aligned. All packages active in requirements.txt and containerized.
Feasibility & Impact	Institutional/Tech/Operational feasibility, 5.7 Cr MSMEs market, Prototype Impact Index chart.	Detailed breakdown of expected outcomes, user effort reduction, and verified live QR code.	Proven by benchmark_evaluation_report.json (9.59/10 accuracy score).

2. End-to-End Codebase Architecture & Technical Execution

The project is structured into an enterprise-grade, modular monorepo spanning full-stack frontend, asynchronous backend services, agentic graph orchestration, and automated regulatory intelligence scrapers.

- **Frontend Interface (React 18 + TypeScript + Vite):** Contains a responsive, accessible interface featuring a public Landing Page with an Interactive Compliance Simulator (Gold Hallmarking, CRS Laptops, Lithium Batteries), Gemini-style multi-turn chat workspace with Server-Sent Events (SSE) token streaming, Dark/Light mode theme engine, live Telemetry Drawer, and an immutable Supabase Audit Ledger modal.
- **FastAPI Gateway & Async REST Microservices (app.py):** Implements OpenAPI/Swagger documentation, GZip response compression, CORS security headers, session history endpoints (/api/v1/sessions), Whisper voice processing (/api/v1/voice-query), user model configuration (BYOK), and ReportLab compliance dossier generation (/api/v1/export/pdf).

- **Agentic StateGraph (src/agent/graph.py & nodes.py):** Orchestrated with LangGraph. Employs zero-token heuristics (`_fast_classify` and `_needs_rewrite`) covering ~90% of routine inquiries without LLM calls. Routes queries dynamically across 7 regulatory domains: hallmark, registration (CRS), certification (Scheme-I/FMCS), laboratory (LRS), manakonline, catalog_search, and chat.
- **Hybrid Dual-Engine Retrieval (src/tools/retriever.py):** Fuses sparse BM25 keyword matching (BM25Okapi) with dense semantic embeddings (ChromaDB / Supabase PGVector). Queries are dispatched concurrently via `ThreadPoolExecutor`, merged, deduplicated, and reranked using cosine similarity.
- **5-Source Live Portal Scraper (src/tools/scraper.py):** Solves the #1 failure mode of traditional RAG (outdated data) by concurrently querying 5 official endpoints: (1) Official BIS 'Know Your Standards' JSON API, (2) Live QCO Gazette Registry, (3) MeitY CRS Electronics Matrix, (4) Conformity Assessment Schemes, and (5) WordPress Gazette Circulars & PDF media attachments.
- **Deterministic Output Guardrails (src/agent/guardrails.py):** Guarantees legal fidelity by normalizing statutory citations under the Bureau of Indian Standards Act, 2016, sanitizing URLs against an official domain whitelist (manakonline.in, crsbis.in, bis.gov.in), and triggering proactive disambiguation workflows when queries are ambiguous.
- **Enterprise BYOK & Multi-Model Engine (src/agent/llm_factory.py):** Supports Google Gemini 3.5 Flash (default), Mistral AI, OpenAI, OpenRouter, and Local Ollama (Llama 3.2, DeepSeek-R1) with automatic deprecation remapping.
- **Immutable Supabase Audit Ledger (backend/db/):** Enforces legal non-repudiation by logging every interaction with SHA-256 prompt/payload hashes, intent confidence scores, and token telemetry in Supabase PostgreSQL.

3. Slide-by-Slide Winning PPT Presentation Speech

This section provides the verbatim presentation script for the 6-slide Smart India Hackathon pitch deck. Each slide contains the key objective, exact timing target, visual/gesture cues, and a word-for-word authoritative speech script.

SLIDE 1: Title Page & Project Mandate | *Target Duration: 0:00 - 0:45 (45s)*

Key Objective: Establish immediate authority, introduce team CONTEXT CREW, and define the critical national mandate of SIH26107.

Visual / Gesture Cue: Stand tall, maintain direct eye contact with the central jury, speak with calm energy, and gesture toward the slide title.

Verbatim Pitch Speech:

Respected Jury Members, Evaluators, and Industry Leaders — good morning! Today, Team CONTEXT CREW is proud to present BIS SATHI, our next-generation, agentic AI assistant engineered specifically for the Bureau of Indian Standards under Problem Statement SIH26107. Every single day, from the helmet strapped on a two-wheeler rider to the gold jewellery in our homes, from consumer electronics to heavy structural steel — quality, safety, and reliability in India are anchored by the Bureau of Indian Standards. Yet, accessing, understanding, and complying with these standards remains one of the largest regulatory bottlenecks for Indian citizens, manufacturers, and our 5.7 Crore MSMEs. We built BIS SATHI to solve this permanently. Let us walk you through how we turned 23,000+ complex standards into instant, actionable, and zero-hallucination conversational intelligence.

SLIDE 2: The Problem & The Solution | *Target Duration: 0:45 - 2:15 (90s)*

Key Objective: Contribute clear problem-solution contrast. Emphasize the 23K standards, 1,437 certified products, MSME pain, and conversational resolution.

Visual / Gesture Cue: Point with left hand to 'THE PROBLEM' panel, pause briefly, then open hands towards 'THE SOLUTION' panel.

Verbatim Pitch Speech:

Look at the left half of this slide: The Real-World Problem. India has over 23,000 active Indian Standards and more than 1,437 product categories under mandatory certification. Where is this information? It is scattered across Manakonline, the CRS portal, gazette notifications, and hundreds of dense PDF circulars. For an everyday consumer, verifying whether gold jewellery or a toy is compliant requires deciphering complex acronyms like HUID, CRS, or STI. For an MSME manufacturer, missing a newly notified Quality Control Order (QCO) deadline can result in factory shipment halts, customs rejection, and severe statutory penalties under Section 29 of the BIS Act, 2016. Conventional search engines only return raw PDFs — they do not guide you through the process. Now look at the right half: The BIS SATHI Solution. BIS SATHI is an Agentic Regulatory Copilot designed for four distinct user classes: Consumers, Industries, MSMEs, and Foreign Importers. The interaction is seamless: A user asks a question via voice or text in English, Hindi, or Hinglish. BIS SATHI understands statutory intent, retrieves exact standards using hybrid vector intelligence, performs live portal checks, and delivers a source-grounded response with clear, actionable next steps and official portal links.

SLIDE 3: Technical Approach & Architecture | *Target Duration: 2:15 - 4:00 (105s)*

Key Objective: Demonstrate technical depth. Explain the dual-path architecture (0-Token Cache vs. Smart AI Path), hybrid retrieval, live scraper, and tech stack.

Visual / Gesture Cue: Trace the workflow from user input to the bifurcation point; highlight '0-Token Fast Path' with emphasis on cost savings.

Verbatim Pitch Speech:

Now, let's look under the hood at our Technical Approach. When building an AI system for government regulatory compliance, two challenges immediately arise: cost scalability and hallucination prevention. We engineered a dual-path workflow to solve both. Every user query enters through our FastAPI Gateway, secured by OAuth and Supabase authentication. Here, the system makes an immediate algorithmic decision: First — The 0-Token Fast Path. If a query matches an already-verified regulatory standard, our SHA-256 multi-key persistent cache answers it in under 50 milliseconds with zero LLM API calls and ■0.00 infrastructure cost. For a national deployment serving millions of citizens asking routine queries like 'What is the hallmark fee?' or 'What is the IS code for packaged water?', this slashes cloud compute expenses by over 80%. Second — The Smart AI Path. For dynamic compliance queries, our LangGraph StateGraph takes over: (1) It rewrites follow-up questions contextually using memory of prior turns. (2) It executes a Parallel Hybrid Search, combining BM25 keyword matching with dense embeddings and cosine reranking. (3) Crucially, it executes a Live 5-Source Portal Check across the official Know Your Standards API, active QCO gazette registries, and MeitY CRS tables. If a Ministry issued an amendment yesterday, BIS SATHI knows it today. (4) Finally, it synthesizes the answer with strict deterministic guardrails that enforce the BIS Act 2016 and whitelist official portal URLs.

SLIDE 4: Feasibility, Viability & Risk Mitigations | Target Duration: 4:00 - 5:30 (90s)

Key Objective: Establish economic, market, and operational viability. Address hallucination and outdated data risks head-on.

Visual / Gesture Cue: Point to the 5.70 Crore MSME stat and the Risk-Mitigation table at the bottom left.

Verbatim Pitch Speech:

Why is BIS SATHI both feasible today and viable tomorrow? From an Institutional Feasibility standpoint, BIS SATHI requires zero disruption to existing BIS infrastructure. It does not replace Manakonline; it sits on top as an intelligent conversational layer, actively driving traffic and valid applications directly into official portals. Operationally, it automates up to 70% of repetitive helpdesk inquiries, freeing BIS technical officers to focus on factory audits and laboratory testing. Now, look at the Economic and Market Viability: India is home to over 5.70 Crore MSMEs and over 51,500 active BIS licenses. With the government rapidly expanding mandatory Quality Control Orders to curb substandard imports, compliance guidance is in unprecedented demand. BIS SATHI follows a scalable lifecycle: from a free citizen-facing consumer tool, to an MSME onboarding portal, to an enterprise compliance engine that compiles audit-ready compliance dossiers. And how do we handle the primary risks? Against AI Hallucination, we employ strict hybrid RAG grounding and deterministic post-synthesis validation. Against Outdated Information, we run live automated scraping of BIS gazettes with in-memory TTL caching. Against Language Barriers, we provide native voice interaction with full support for Hindi, Hinglish, and English.

SLIDE 5: Impact, Benefits & Impact Index | Target Duration: 5:30 - 6:45 (75s)

Key Objective: Present the 4 impact pillars (Citizens, Industry, Services, Digital India) and quantify user effort reduction using the Impact Index.

Visual / Gesture Cue: Emphasize the contrast bars in the Impact Index chart; gesture upward when discussing Digital India.

Verbatim Pitch Speech:

The impact of BIS SATHI aligns directly with the Digital India vision across four foundational pillars: First, Empowering Citizens: An everyday consumer can ask about gold jewellery purity or toy safety in their native language and receive instant, verifiable answers. Second, Supporting Industry and MSMEs: Instead of spending weeks hiring costly compliance consultants to identify applicable IS standards, an entrepreneur receives an exact 8-step roadmap, required lab testing parameters, and applicable fee concessions in 10 seconds. Third, Enhancing BIS Operations: By resolving routine tier-1 queries automatically, the Bureau's administrative backlog is drastically reduced. And fourth, National Economic Growth: Facilitating faster certifications directly accelerates domestic manufacturing under 'Make in India' and prevents non-compliant foreign goods from entering our domestic supply chain. As shown in our Prototype Impact Index, compared to traditional multi-portal manual search, BIS SATHI reduces user research effort by over 85%, while elevating actionable guidance and standards discovery to near-perfect scores.

SLIDE 6: Research, References & Live Demo Transition | Target Duration: 6:45 - 7:30 (45s)

Key Objective: Cite authoritative research, latest regulatory news context, show UI preview, and transition into the live interactive demonstration.

Visual / Gesture Cue: Present the QR code on screen; invite jury to scan with smartphones while switching screen to live web application.

Verbatim Pitch Speech:

Our system is grounded in authoritative sources: the official Bureau of Indian Standards portals, the Bureau of Indian Standards Act 2016, MeitY gazette notifications, and real-time regulatory circulars. We have tracked latest developments — including high-tech industry BIS exemptions, unified digital modernization under ManakOnline, and shifting QCO enforcement deadlines — to ensure our knowledge base reflects current statutory reality. Most importantly — this is not a conceptual mockup or a Figma prototype. BIS SATHI is completely built, containerized, and deployed live right now. You can scan the QR code on your screens or visit bis-agnet.app to test it in real time. With the Jury's permission, we would now like to demonstrate BIS SATHI live in action!

4. The 2-Minute Live Demonstration Playbook

Execute this precise step-by-step choreography during the live evaluation to showcase real-time voice recognition, zero-token cache economics, deep compliance research, PDF generation, and tamper-evident audit logging.

Step 1: Public Landing Page & Interactive Simulator

Execution Time: 0:00 - 0:25

Choreography & Speech: Open <https://bis-agnet.app/>. Highlight that unauthenticated visitors have immediate access to an interactive regulatory simulator. Click 'Gold Hallmarking' or 'Electronics CRS' to showcase instant standard extraction (IS 1417 / IS 13252), statutory gazette citations, and verified key points without login friction.

Step 2: Multilingual Voice Input via Whisper

Execution Time: 0:25 - 0:55

Choreography & Speech: Log in. Click the Microphone icon. Speak in natural Hinglish: 'Kya mobile charger ke liye BIS CRS registration lena zaroori hai?'. Show the real-time SSE token stream. Highlight that the system automatically matched Hinglish input style, cited IS 13252 (Part 1), detailed NABL test report rules, and linked to crsbis.in.

Step 3: Zero-Token Instant Cache Hit & Telemetry

Execution Time: 0:55 - 1:20

Choreography & Speech: Open the Telemetry Drawer on the right. Ask a common query: 'What is the procedure for getting a Gold Hallmarking license under BIS?'. Show the Telemetry panel: Latency = 42ms, Tokens = 0, Cache Status = ■ 0-Token Instant Cache Hit. Explain how this slashes national cloud hosting costs to ■ 0.00.

Step 4: Deep Research Mode & Publication-Grade PDF Export

Execution Time: 1:20 - 1:45

Choreography & Speech: Enter: 'Generate a detailed compliance research dossier for IS 12701 polyethylene water tanks'. The system executes deep research, pulling chemical migration limits (60 mg/L) and capacity limits (10,000L). Click 'Download Research PDF'. Open the generated PDF to show the official 13-section BIS dossier with running headers/footers.

Step 5: Tamper-Evident Supabase Audit Ledger

Execution Time: 1:45 - 2:00

Choreography & Speech: Click 'Audit Ledger' in the top navigation. Show the admin table in Supabase PostgreSQL: every turn is recorded with SHA-256 prompt/payload hashes, intent confidence scores, and token usage, proving complete legal non-repudiation.

5. Jury Defense Q&A; Battlecard

Anticipated high-stakes technical, legal, and business questions prepared for the SIH Jury with sharp, winning answers.

Q1: "How do you guarantee that your AI will not hallucinate a fake Indian Standard (IS Code) or incorrect statutory penalty?"

Strategic Defense Angle: Three-Layer Defense: Grounding, Deterministic Guardrails, Source Verifiability

Winning Answer:

We eliminate hallucinations through a three-layer defense: (1) Strict Grounding: The agent synthesizes answers strictly from retrieved documents and live official API records. (2) Deterministic Post-Synthesis Guardrails (src/agent/guardrails.py): Output is validated via regex for valid standard formats (IS \d+), non-governmental links are purged, and penalties are legally locked to the Bureau of Indian Standards Act, 2016. (3) Source Transparency: Every response cites official document sources and portal links for instant human verification.

Q2: "BIS standards and Quality Control Orders update frequently. How do you keep the database fresh without re-embedding 23,000 PDFs daily?"

Strategic Defense Angle: 5-Source Real-Time Scraper + In-Memory TTL Cache

Winning Answer:

Static vector databases become obsolete quickly. Our architecture pairs foundational vector collections with an active 5-Source Live Regulatory Intelligence Scraper (src/tools/scrapper.py). It queries the official 'Know Your Standards' JSON API, upcoming QCO gazette registries, and MeitY CRS tables concurrently on demand. When a Ministry notifies a new deadline, our live scraper retrieves it within seconds, bypassing the need for expensive daily vector re-indexing.

Q3: "What makes this an 'Agentic AI' rather than a standard RAG pipeline?"

Strategic Defense Angle: Autonomous State Machine, Contextual Query Rewriter, Active Disambiguation

Winning Answer:

A basic RAG bot takes a prompt, embeds it, and prints text. BIS SATHI is an Agentic StateGraph built on LangGraph: (1) It dynamically classifies intent across 7 regulatory domains. (2) It self-corrects and rewrites short replies ('yes, tell me fees') into fully qualified queries based on conversation memory. (3) It detects ambiguous queries like 'pipe' and actively triggers a multi-choice disambiguation prompt (HDPE vs UPVC vs Steel). (4) It autonomously invokes live web scrapers, persistent caches, and ReportLab PDF compilers.

Q4: "What is the cost economics of deploying this nationally for 5.7 Crore MSMEs?"

Strategic Defense Angle: 0-Token Fast Path + Zero-Token Intent Heuristics + BYOK/Ollama Sovereign Deployments

Winning Answer:

Cost scalability was our core design priority: (1) Through our 0-Token Fast Path, repetitive routine inquiries (over 80% of volume) are served from our SHA-256 persistent cache at ₹0.00 LLM cost. (2) Zero-token regex heuristics classify greetings and IS codes without LLM invocation. (3) For dynamic turns, Gemini 3.5 Flash and Mistral deliver sub-second latency at fractional token costs. (4) For sovereign government deployment, our BYOK and Ollama engine allows running open-source models (Llama 3.2, DeepSeek-R1) completely on on-premises hardware with zero recurring API expenses.

Q5: "How does BIS SATHI serve non-English speakers and small artisans who cannot read complex English technical guidelines?"

Strategic Defense Angle: Whisper Speech-to-Text + Strict Script Matching in Hindi and Hinglish

Winning Answer:

Accessibility is fundamental: (1) Integrated Whisper voice recognition lets users speak directly into their mobile phones. (2) Strict Script & Language Matching ensures that queries in Hindi receive replies in formal Devanagari Hindi, while queries in Hinglish receive natural Romanized Hinglish. (3) In our empirical benchmark evaluation (data/benchmark_evaluation_report.json), Hindi queries achieved an outstanding accuracy score of 9.6 out of 10, correctly identifying Form H-1 hallmarking registration steps and exemption limits.

Q6: "How easily can BIS SATHI integrate with existing BIS systems like ManakOnline and e-BIS?"

Strategic Defense Angle: Containerized Microservice + Documented REST Endpoints + Non-Disruptive Overlay

Winning Answer:

BIS SATHI is 100% cloud-agnostic and container-ready with our provided Dockerfile and docker-compose configurations. It exposes documented FastAPI REST endpoints with OpenAPI specs that can be embedded into Manakonline, the BIS Care mobile app, or national citizen portals like India.gov.in as a single microservice with zero architectural overhaul required.

6. Project Technical Specifications & Evaluation Summary

A consolidated summary of core technical attributes, performance benchmarks, and deployment parameters.

Metric / Parameter	Specification & Implemented Value	Architectural Advantage
Problem Statement	SIH26107 (Smart Automation Software)	Direct alignment with official Ministry mandate for BIS
Average Accuracy	9.59 / 10 (Empirically tested across Easy/Med/Complex queries)	Verified compliance metadata, exact IS codes, and portal URLs
Turn Latency (Cache)	< 50 ms (Sub-second instant response)	0 LLM API calls, ₹0.00 compute cost on repeated queries
Agent Framework	LangGraph StateGraph with conditional routing	Deterministic multi-turn context preservation and self-correction
Retrieval Architecture	Parallel Hybrid: BM25 Okapi + Dense Vector + Cosine Reranking	High recall keyword matching combined with semantic nuance
Live Portal Scraper	5-Source Concurrent: Know Your Standards API + QCOs + CRS	Zero lag on newly notified Gazette amendments and deadlines
Guardrails Layer	Rule-based BIS Act 2016 enforcement & portal URL whitelisting	Complete elimination of hallucinated acts and unauthorized URLs
Supported LLMs	Gemini 3.5 Flash, Mistral, OpenAI, OpenRouter, Local Ollama	Multi-provider BYOK + Sovereign zero-cloud deployment option
Voice Processing	OpenAI Whisper (offline / low latency)	Hands-free vernacular accessibility for artisans and MSMEs
Compliance Export	ReportLab 13-Section PDF Regulatory Dossier Generator	Instant printable audit document for factory compliance officers
Audit & Security	Supabase PostgreSQL with SHA-256 immutable interaction hashing	Tamper-evident legal non-repudiation for regulatory audits
Live Deployment	Frontend on Vercel, Backend on Render/Docker, DB on Supabase	Publicly accessible worldwide at https://bis-agnet.app/

PRESENTATION READINESS CONFIRMATION

Summary for the Presentation Team: With verified 9.59/10 accuracy, sub-50ms cache response, live 5-source regulatory scraping, publication-grade PDF compilation, and an active production deployment at <https://bis-agnet.app/>, BIS SATHI represents a complete, robust, and winning solution for SIH26107.